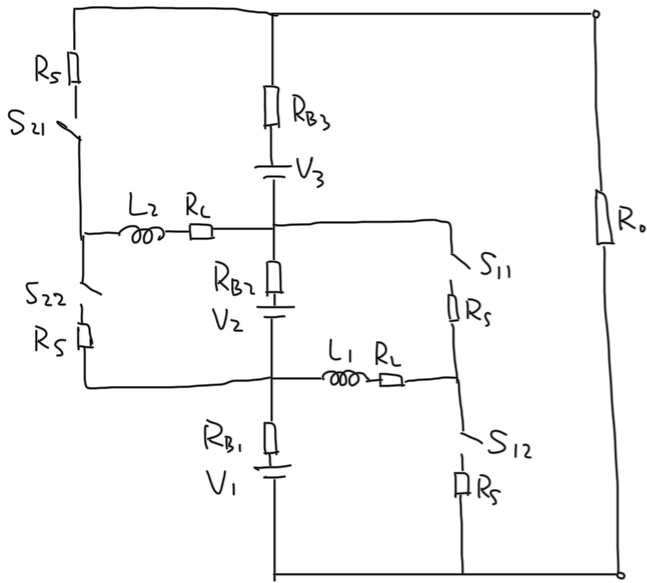
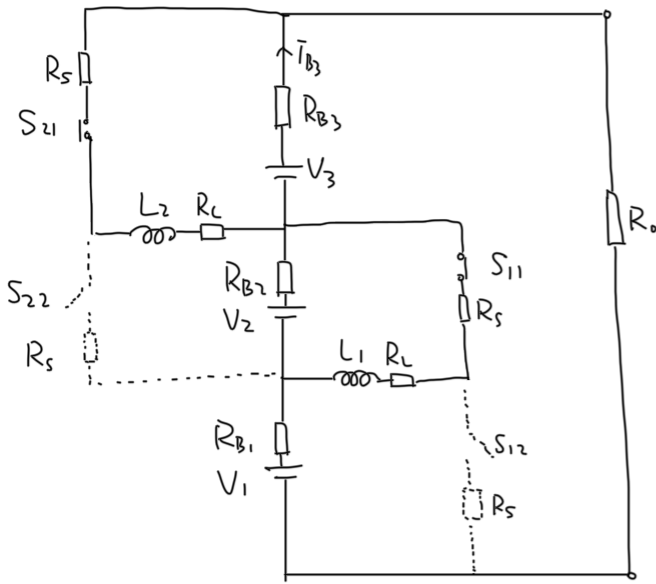
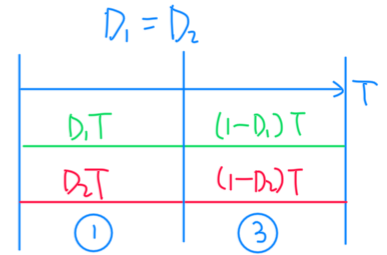
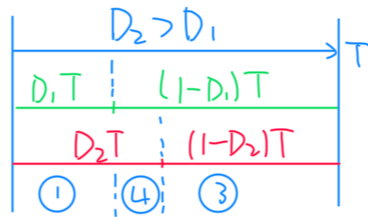
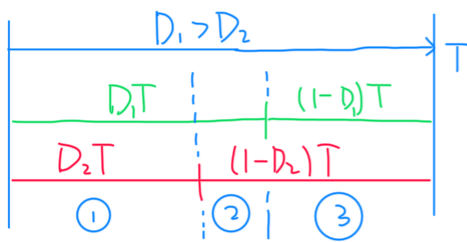




All Four cases:



D_1T & D_2T $[\dot{i}_{L1}, \dot{i}_{L2}] = x$

$$\begin{cases} V_2 = \bar{i}_{B2} R_{B2} + \dot{i}_{L1} (R_5 + R_L) + L_1 \frac{d\dot{i}_{L1}}{dt} \\ V_3 = \bar{i}_{B3} R_{B3} + \dot{i}_{L2} (R_5 + R_L) + L_2 \frac{d\dot{i}_{L2}}{dt} \end{cases}$$

$$\begin{cases} \frac{d\dot{i}_{L1}}{dt} = -\frac{R_5 + R_L}{L_1} \dot{i}_{L1} + \frac{1}{L_1} V_2 - \frac{R_{B1}}{L_1} \bar{i}_{B2} \\ \frac{d\dot{i}_{L2}}{dt} = -\frac{R_5 + R_L}{L_2} \dot{i}_{L2} + \frac{1}{L_2} V_3 - \frac{R_{B3}}{L_2} \bar{i}_{B3} \end{cases}$$

$$\bar{i}_{B_1} + \bar{i}_{L_1} = \bar{i}_{B_2} \quad \bar{i}_{B_1} = \bar{i}_{B_2} - \bar{i}_{L_1}$$

$$\bar{i}_{B_2} + \bar{i}_{L_2} = \bar{i}_{B_3} + \bar{i}_{L_1} \quad \bar{i}_{B_2} = \bar{i}_{B_3} + \bar{i}_{L_1} - \bar{i}_{L_2}$$

$$\bar{i}_{B_3} = \bar{i}_{L_2} + \bar{i}_{\text{pack}}$$

$$V_1 + V_2 + V_3 = \bar{i}_{B_1} R_{B_1} + \bar{i}_{B_2} R_{B_2} + \bar{i}_{B_3} R_{B_3} + \bar{i}_{\text{pack}} R_o$$

$$V_1 + V_2 + V_3 = (\bar{i}_{B_3} - \bar{i}_{L_2}) R_{B_1} + (\bar{i}_{B_3} + \bar{i}_{L_1} - \bar{i}_{L_2}) R_{B_2} + \bar{i}_{B_3} R_{B_3} + (\bar{i}_{B_3} - \bar{i}_{L_2}) R_o$$

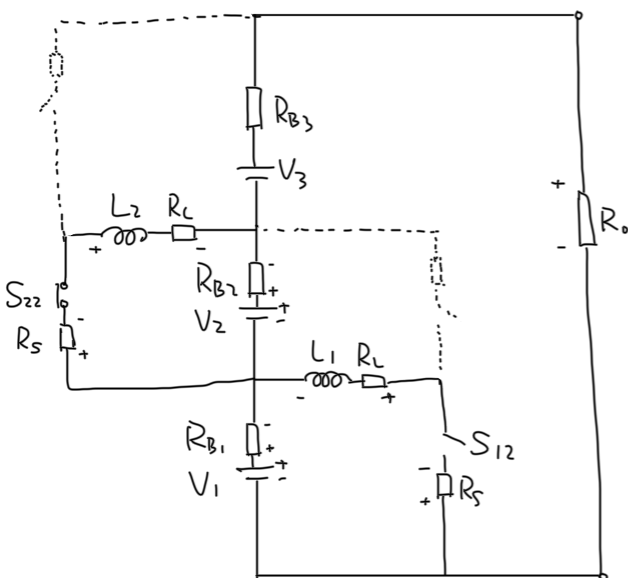
$$V_1 + V_2 + V_3 + \bar{i}_{L_2} R_{B_1} - \bar{i}_{L_1} R_{B_2} + \bar{i}_{L_2} R_{B_2} + \bar{i}_{L_2} R_o = \bar{i}_{B_3} (R_{B_1} + R_{B_2} + R_{B_3} + R_o)$$

$$\left[\begin{aligned} \bar{i}_{B_3} &= \frac{V_1 + V_2 + V_3 - R_{B_2} \bar{i}_{L_1} + (R_{B_1} + R_{B_2} + R_o) \bar{i}_{L_2}}{R_{B_1} + R_{B_2} + R_{B_3} + R_o} = \frac{V_1 + V_2 + V_3}{R_{B_1} + R_{B_2} + R_{B_3} + R_o} - \frac{R_{B_2} (\bar{i}_{L_1})}{R_{B_1} + R_{B_2} + R_{B_3} + R_o} + \frac{(R_{B_1} + R_{B_2} + R_o) (\bar{i}_{L_2})}{R_{B_1} + R_{B_2} + R_{B_3} + R_o} \\ \bar{i}_{B_2} &= \bar{i}_{B_3} + \bar{i}_{L_1} - \bar{i}_{L_2} = \frac{V_1 + V_2 + V_3}{\Sigma R} + \frac{\Sigma R - R_{B_2}}{\Sigma R} \bar{i}_{L_1} - \frac{R_{B_3}}{\Sigma R} \bar{i}_{L_2} \quad (\Sigma R = R_{B_1} + R_{B_2} + R_{B_3} + R_o) \\ \bar{i}_{B_1} &= \bar{i}_{B_3} - \bar{i}_{L_2} \end{aligned} \right.$$

$$\left[\begin{aligned} \frac{di_{L_1}}{dt} &= -\frac{R_s + R_L}{L_1} \bar{i}_{L_1} + \frac{1}{L_1} V_2 - \frac{R_{B_1}}{L_1} \left[\frac{V_1 + V_2 + V_3}{\Sigma R} + \frac{\Sigma R - R_{B_2}}{\Sigma R} \bar{i}_{L_1} - \frac{R_{B_3}}{\Sigma R} \bar{i}_{L_2} \right] \\ \frac{di_{L_2}}{dt} &= -\frac{R_s + R_L}{L_2} \bar{i}_{L_2} + \frac{1}{L_2} V_3 - \frac{R_{B_3}}{L_2} \left[\frac{V_1 + V_2 + V_3}{R_{B_1} + R_{B_2} + R_{B_3} + R_o} - \frac{R_{B_2} (\bar{i}_{L_1})}{R_{B_1} + R_{B_2} + R_{B_3} + R_o} + \frac{(R_{B_1} + R_{B_2} + R_o) (\bar{i}_{L_2})}{R_{B_1} + R_{B_2} + R_{B_3} + R_o} \right] \end{aligned} \right.$$

$$\left[\begin{aligned} \dot{\bar{i}}_{L_1} &= -\left[\frac{(\Sigma R - R_{B_2}) R_{B_2}}{L_1 \Sigma R} + \frac{R_s + R_L}{L_1} \right] \bar{i}_{L_1} + \frac{R_{B_2} R_{B_3}}{L_1 \Sigma R} \bar{i}_{L_2} - \frac{R_{B_2}}{L_1 \Sigma R} V_1 + \frac{\Sigma R - R_{B_1}}{L_1 \Sigma R} V_2 - \frac{R_{B_2}}{L_1 \Sigma R} V_3 \end{aligned} \right.$$

$$\left[\begin{aligned} \dot{\bar{i}}_{L_2} &= -\left[\frac{(R_{B_1} + R_{B_2} + R_o) R_{B_3}}{(R_{B_1} + R_{B_2} + R_{B_3} + R_o) L_2} + \frac{R_s + R_L}{L_2} \right] \bar{i}_{L_2} + \frac{R_{B_2} R_{B_3}}{(R_{B_1} + R_{B_2} + R_{B_3} + R_o) L_2} \bar{i}_{L_1} - \frac{R_{B_3}}{L_2 \Sigma R} V_1 - \frac{R_{B_3}}{L_2 \Sigma R} V_2 + \frac{\Sigma R - R_{B_3}}{L_2 \Sigma R} V_3 \end{aligned} \right.$$



$$(1-D_1)T \& (1-D_2)T \quad \bar{i}_{L_1} \quad \bar{i}_{L_2}$$

$$\begin{cases} \bar{i}_{B1} R_{B1} = V_1 + L_1 \frac{di_{L1}}{dt} + \bar{i}_{L1} (R_S + R_L) \\ \bar{i}_{B2} R_{B2} = V_2 + L_2 \frac{di_{L2}}{dt} + \bar{i}_{L2} (R_S + R_L) \\ \dot{\bar{i}}_{L1} = -\frac{R_S + R_L}{L_1} \bar{i}_{L1} - \frac{1}{L_1} V_1 + \frac{R_{B1}}{L_1} \bar{i}_{B1} \\ \dot{\bar{i}}_{L2} = -\frac{R_S + R_L}{L_2} \bar{i}_{L2} - \frac{1}{L_2} V_2 + \frac{R_{B2}}{L_2} \bar{i}_{B2} \end{cases}$$

$$\begin{cases} \bar{i}_{B1} + \bar{i}_{L2} = \bar{i}_{B3} \\ \bar{i}_{B1} + \bar{i}_{L1} = \bar{i}_{B2} + \bar{i}_{L1} \\ V_1 + V_2 + V_3 = \bar{i}_{B1} R_{B1} + \bar{i}_{B2} R_{B2} + \bar{i}_{B3} (R_{B3} + R_0) \end{cases} \quad \begin{cases} \bar{i}_{B2} = \bar{i}_{B3} - \bar{i}_{L2} \\ \bar{i}_{B1} = \bar{i}_{B3} - \bar{i}_{L1} \end{cases}$$

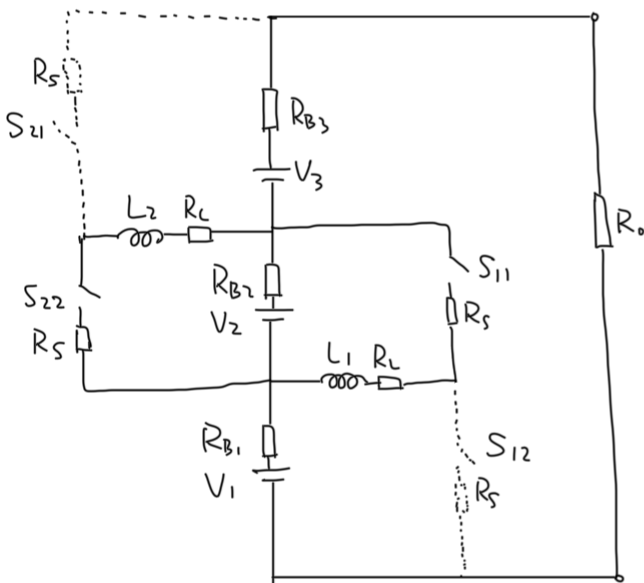
$$V_1 + V_2 + V_3 = (\bar{i}_{B3} - \bar{i}_{L1}) R_{B1} + (\bar{i}_{B3} - \bar{i}_{L2}) R_{B2} + \bar{i}_{B3} (R_{B3} + R_0)$$

$$V_1 + V_2 + V_3 + R_{B1} \bar{i}_{L1} + R_{B2} \bar{i}_{L2} = (R_{B1} + R_{B2} + R_{B3} + R_0) \bar{i}_{B3}$$

$$\begin{cases} \bar{i}_{B3} = \frac{V_1 + V_2 + V_3 + R_{B1} \bar{i}_{L1} + R_{B2} \bar{i}_{L2}}{\Sigma R} \\ \bar{i}_{B2} = \frac{V_1 + V_2 + V_3 + R_{B1} \bar{i}_{L1} - (\Sigma R - R_{B2}) \bar{i}_{L2}}{\Sigma R} \\ \bar{i}_{B1} = \frac{V_1 + V_2 + V_3 - (\Sigma R - R_{B1}) \bar{i}_{L1} + R_{B2} \bar{i}_{L2}}{\Sigma R} \end{cases}$$

$$\begin{cases} \dot{\bar{i}}_{L1} = -\frac{R_S + R_L}{L_1} \bar{i}_{L1} - \frac{1}{L_1} V_1 + \frac{R_{B1}}{L_1} \frac{V_1 + V_2 + V_3 - (\Sigma R - R_{B1}) \bar{i}_{L1} + R_{B2} \bar{i}_{L2}}{\Sigma R} \\ \dot{\bar{i}}_{L2} = -\frac{R_S + R_L}{L_2} \bar{i}_{L2} - \frac{1}{L_2} V_2 + \frac{R_{B2}}{L_2} \frac{V_1 + V_2 + V_3 + R_{B1} \bar{i}_{L1} - (\Sigma R - R_{B2}) \bar{i}_{L2}}{\Sigma R} \end{cases}$$

$$\begin{cases} \dot{\bar{i}}_{L1} = -\left[\frac{(R_S + R_L) \Sigma R + R_{B1} (\Sigma R - R_{B1})}{L_1 \Sigma R} \right] \bar{i}_{L1} + \frac{R_{B1} R_{B2}}{L_1 \Sigma R} \bar{i}_{L2} + \frac{R_{B1} - \Sigma R}{L_1 \Sigma R} V_1 + \frac{R_{B1}}{L_1 \Sigma R} V_2 + \frac{R_{B1}}{L_1 \Sigma R} V_3 \\ \dot{\bar{i}}_{L2} = \frac{R_{B2} R_{B1}}{L_2 \Sigma R} \bar{i}_{L1} - \left[\frac{\Sigma R (R_S + R_L) + R_{B2} (\Sigma R - R_{B2})}{L_2 \Sigma R} \right] \bar{i}_{L2} + \frac{R_{B2}}{L_2 \Sigma R} V_1 + \frac{R_{B2} - \Sigma R}{L_2 \Sigma R} V_2 + \frac{R_{B2}}{L_2 \Sigma R} V_3 \end{cases}$$



$D_1 T \& (1-D_2) T \quad i_{L1} \quad i_{L2}$

$$\begin{cases} V_2 = R_{B2} \bar{i}_{B2} + (R_S + R_L) \bar{i}_{L1} + L_1 \frac{di_{L1}}{dt} \\ R_{B2} \bar{i}_{B2} = V_2 + (R_S + R_L) \bar{i}_{L2} + L_2 \frac{di_{L2}}{dt} \end{cases}$$

$$\begin{cases} \bar{i}_{B1} + \bar{i}_{L1} = \bar{i}_{B2} + \bar{i}_{L2} \\ \bar{i}_{L2} + \bar{i}_{B2} = \bar{i}_{B3} + \bar{i}_{L1} \\ V_1 + V_2 + V_3 = R_{B1}\bar{i}_{B1} + R_{B2}\bar{i}_{B2} + R_{B3}\bar{i}_{B3} + R_o\bar{i}_{B3} \end{cases}$$

$$\begin{cases} \bar{i}_{B1} = \bar{i}_{B2} + \bar{i}_{L2} - \bar{i}_{L1} = \bar{i}_{B3} \\ \bar{i}_{B2} = \bar{i}_{B3} + \bar{i}_{L1} - \bar{i}_{L2} \end{cases}$$

$$V_1 + V_2 + V_3 = R_{B1}\bar{i}_{B3} + R_{B2}(\bar{i}_{B3} + \bar{i}_{L1} - \bar{i}_{L2}) + (R_{B3} + R_o)\bar{i}_{B3}$$

$$V_1 + V_2 + V_3 - R_{B2}\bar{i}_{L1} + R_{B2}\bar{i}_{L2} = R_{B1}\bar{i}_{B3} + R_{B2}\bar{i}_{B3} + (R_{B3} + R_o)\bar{i}_{B3}$$

$$\bar{i}_{B3} = \frac{V_1 + V_2 + V_3 - R_{B2}\bar{i}_{L1} + R_{B2}\bar{i}_{L2}}{\Sigma R}$$

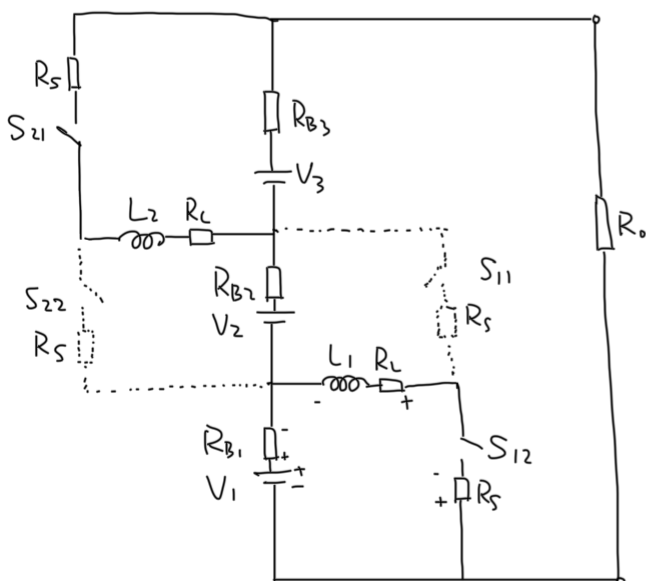
$$\bar{i}_{B2} = \frac{V_1 + V_2 + V_3 + (\Sigma R - R_{B2})\bar{i}_{L1} + (R_{B2} - \Sigma R)\bar{i}_{L2}}{\Sigma R}$$

$$\begin{cases} V_2 = R_{B2}\bar{i}_{B2} + (R_s + R_L)\bar{i}_{L1} + L_1 \frac{d\bar{i}_{L1}}{dt} \\ R_{B2}\bar{i}_{B2} = V_2 + (R_s + R_L)\bar{i}_{L2} + L_2 \frac{d\bar{i}_{L2}}{dt} \end{cases}$$

$$\dot{\bar{i}}_{L1} = -R_{B2} \frac{V_1 + V_2 + V_3 + (\Sigma R - R_{B2})\bar{i}_{L1} + (R_{B2} - \Sigma R)\bar{i}_{L2}}{L_1 \Sigma R} - \frac{R_s + R_L}{L_1} \bar{i}_{L1} + \frac{1}{L_1} V_2$$

$$\dot{\bar{i}}_{L2} = R_{B2} \frac{V_1 + V_2 + V_3 + (\Sigma R - R_{B2})\bar{i}_{L1} + (R_{B2} - \Sigma R)\bar{i}_{L2}}{L_2 \Sigma R} - \frac{R_s + R_L}{L_2} \bar{i}_{L2} - \frac{1}{L_2} V_2$$

$$\begin{cases} \dot{\bar{i}}_{L1} = - \left[\frac{R_{B2}(\Sigma R - R_{B2}) + \Sigma R \cdot (R_s + R_L)}{L_1 \Sigma R} \right] \bar{i}_{L1} - \frac{R_{B2}(R_{B2} - \Sigma R)}{L_1 \Sigma R} \bar{i}_{L2} - \frac{R_{B2}}{L_1 \Sigma R} V_1 + \frac{\Sigma R - R_{B2}}{L_1 \Sigma R} V_2 - \frac{R_{B2}}{L_1 \Sigma R} V_3 \\ \dot{\bar{i}}_{L2} = \frac{R_{B2}(\Sigma R - R_{B2})}{L_2 \Sigma R} \bar{i}_{L1} + \frac{R_{B2}(R_{B2} - \Sigma R) - \Sigma R(R_s + R_L)}{L_2 \Sigma R} \bar{i}_{L2} + \frac{R_{B2}}{L_2 \Sigma R} V_1 + \frac{R_{B2} - \Sigma R}{L_2 \Sigma R} V_2 + \frac{R_{B2}}{L_2 \Sigma R} V_3 \end{cases}$$



(1-D1)T & D2T $\bar{i}_{L1}$ $\bar{i}_{L2}$

$$\begin{cases} V_3 = R_{B_3} \dot{i}_{B_3} + (R_s + R_L) \dot{i}_{L_2} + L_2 \frac{d\dot{i}_{L_2}}{dt} \\ R_{B_1} \dot{i}_{B_1} = V_1 + (R_s + R_L) \dot{i}_{L_1} + L_1 \frac{d\dot{i}_{L_1}}{dt} \end{cases}$$

$$\begin{cases} \dot{i}_{B_1} + \dot{i}_{L_1} = \dot{i}_{B_2} \\ \dot{i}_{B_2} + \dot{i}_{L_2} = \dot{i}_{B_3} \\ \dot{i}_{B_3} = \dot{i}_{L_2} + \dot{i}_{pack} \\ V_1 + V_2 + V_3 = R_{B_1} \dot{i}_{B_1} + R_{B_2} \dot{i}_{B_2} + R_{B_3} \dot{i}_{B_3} + R_o \dot{i}_{pack} \end{cases} \quad \begin{cases} \dot{i}_{pack} = \dot{i}_{B_2} \\ \dot{i}_{B_3} = \dot{i}_{L_2} + \dot{i}_{B_2} \\ \dot{i}_{B_1} = \dot{i}_{B_2} - \dot{i}_{L_1} \end{cases}$$

$$V_1 + V_2 + V_3 = R_{B_1} (\dot{i}_{B_2} - \dot{i}_{L_1}) + R_{B_2} \dot{i}_{B_2} + R_{B_3} (\dot{i}_{L_2} + \dot{i}_{B_2}) + R_o \dot{i}_{B_2}$$

$$V_1 + V_2 + V_3 + R_{B_1} \dot{i}_{L_1} - R_{B_3} \dot{i}_{L_2} = (R_{B_1} + R_{B_2} + R_{B_3} + R_o) \dot{i}_{B_2}$$

$$\dot{i}_{B_2} = \frac{V_1 + V_2 + V_3 + R_{B_1} \dot{i}_{L_1} - R_{B_3} \dot{i}_{L_2}}{\Sigma R}$$

$$\begin{cases} \dot{i}_{B_3} = \frac{V_1 + V_2 + V_3 + R_{B_1} \dot{i}_{L_1} + (\Sigma R - R_{B_3}) \dot{i}_{L_2}}{\Sigma R} \\ \dot{i}_{B_1} = \frac{V_1 + V_2 + V_3 + (R_{B_1} - \Sigma R) \dot{i}_{L_1} - R_{B_3} \dot{i}_{L_2}}{\Sigma R} \end{cases}$$

$$\begin{cases} V_3 = R_{B_3} \dot{i}_{B_3} + (R_s + R_L) \dot{i}_{L_2} + L_2 \frac{d\dot{i}_{L_2}}{dt} \\ R_{B_1} \dot{i}_{B_1} = V_1 + (R_s + R_L) \dot{i}_{L_1} + L_1 \frac{d\dot{i}_{L_1}}{dt} \end{cases}$$

$$\begin{cases} \dot{i}_{L_1} = R_{B_1} \frac{V_1 + V_2 + V_3 + (R_{B_1} - \Sigma R) \dot{i}_{L_1} - R_{B_3} \dot{i}_{L_2}}{L_1 \Sigma R} - \frac{R_s + R_L}{L_1} \dot{i}_{L_1} - \frac{1}{L_1} V_1 \\ \dot{i}_{L_2} = -R_{B_3} \frac{V_1 + V_2 + V_3 + R_{B_1} \dot{i}_{L_1} + (\Sigma R - R_{B_3}) \dot{i}_{L_2}}{L_2 \Sigma R} - \frac{R_s + R_L}{L_2} \dot{i}_{L_2} + \frac{1}{L_2} V_3 \end{cases}$$

$$\begin{cases} \dot{i}_{L_1} = \left[\frac{R_{B_1}(R_{B_1} - \Sigma R) - \Sigma R(R_s + R_L)}{L_1 \Sigma R} \right] \dot{i}_{L_1} - \frac{R_{B_1} R_{B_3}}{L_1 \Sigma R} \dot{i}_{L_2} + \frac{R_{B_1} - \Sigma R}{L_1 \Sigma R} V_1 + \frac{R_{B_1}}{L_1 \Sigma R} V_2 + \frac{R_{B_1}}{L_1 \Sigma R} V_3 \\ \dot{i}_{L_2} = -\frac{R_{B_3} R_{B_1}}{L_2 \Sigma R} \dot{i}_{L_1} + \left[\frac{R_{B_3}(R_{B_3} - \Sigma R) - \Sigma R(R_s + R_L)}{L_2 \Sigma R} \right] \dot{i}_{L_2} - \frac{R_{B_3}}{L_2 \Sigma R} V_1 - \frac{R_{B_3}}{L_2 \Sigma R} V_2 + \frac{\Sigma R - R_{B_3}}{L_2 \Sigma R} V_3 \end{cases}$$